Shifting Stories, Shifting Returns

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https://github.com/komalniraula/Shifting-Stories--Shifting-Returns.git

Summary

This study investigates how longitudinal changes in corporate narratives influence subsequent stock returns, highlighting their relevance as a complementary source of predictive information beyond traditional quantitative factors. Building on the growing field of financial text analysis and narrative disclosure, the research leverages quarterly financial reports and earnings call transcripts from the Orbit dataset to construct a comprehensive feature set capturing sentiment, financial and business content, risk disclosure, subjectivity, and temporal framing.

Multiple Moving Target Score (MTS) variants were developed to quantify quarter-over-quarter shifts across these features. Smoother and weighted formulations (rolling, EWMA, median, variance, frequency) produced unstable or negligible returns, while machine-learning methods (PCA, Lasso) were inconsistent. In contrast, distance-based measures such as KL divergence and cosine similarity yielded the most robust performance, achieving Sharpe ratios up to 0.54 with 75% hit rates. Factor regressions further revealed that while beta exposures to market, value, and momentum are often substantial, positive alpha arises when narrative signals capture information not already explained by these standard risk factors. A composite strategy using principal component analysis (PCA) to blend cosine and variance-based signals achieved improved robustness and cumulative returns.

1 Relevance

Textual data have increasingly been used to predict company performance and stock returns. The analysis of market sentiment, management tone, and disclosure statements has shown consistent forecasting power across settings. Understanding the narratives in financial communications has become increasingly important in both research and practice. Unlike purely quantitative metrics, these narratives can reveal hard-to-quantify aspects of firm fundamentals that other disclosures miss. The growing field of financial narrative processing underscores this importance, with surging interest in applying NLP to corporate reports and using state-of-the-art methods (e.g., word embeddings and transformer-based entailment models) to extract insights from qualitative text.

A particularly relevant insight is that changes in a company's narrative over time can be even more informative than static text. When management's tone or emphasis shifts from one quarter to the next, it often signals a change in underlying outlook or conditions. Studies have shown that tone changes carry significant predictive power. A quarter-over-quarter increase in negativity in earnings calls, termed a "bleak tone change," is regarded as a strong signal of lower future earnings and heightened uncertainty for the firm. Notably, the market tends to initially underreact to these narrative shifts: stock prices move in the expected direction but persistently, indicating that the full implications are not immediately priced in (**druz2020**). This underreaction suggests an opportunity for investors to benefit by paying attention to narrative changes. Indeed, broader evidence shows that shifts in narratives can be exploited: one recent study constructed "narrative momentum" portfolios and found that stocks associated with rising narrative themes outperformed those with declining themes by about 8% annually, delivering alpha beyond standard risk factors (**ssrn2024**).

The "Moving Targets" paper from Harvard (**cohen2020**) adds a critical perspective. The study highlights that managers often alter their KPI framing in response to weakening fundamentals, and that these narrative substitutions are not random. This provides an academic foundation for using longitudinal textual analysis as a leading indicator of risk. All these findings highlight that when corporate storytelling changes, it often foreshadows real outcomes and market movements that systematic analysis can capture.

This work builds on these insights by identifying narrative shifts in quarterly reports and quantifying them for use in a trading strategy. Based on previous research, multiple dimensions of the narrative are examined to capture a comprehensive picture of how it evolves. In particular, this research incorporates: tone (sentiment), financial and business content relevance, risk-related disclosures, analyst mentions and discussion, subjective vs. objective phrasing, and forward-looking vs. historical orientation of each page.

By combining these features, the approach provides a rich, multidimensional view of narrative change. This is highly relevant in today's markets because it merges financial domain knowledge with cutting-edge NLP techniques, enabling systematic detection of subtle shifts in managerial emphasis. As shown later in the methodology and findings sections, such shifts can be transformed into a tradable signal.

2 Methodology

The methodology consisted of three main stages: data preparation, feature construction, and signal development with benchmarking and robustness checks.

2.1 Data Preparation

Corporate filings were obtained from the Orbit dataset, with parsed `blocks` and `pages` files. Each report was linked to factor data and aligned with historical stock returns from CRSP.

Positions were entered on the first trading day of the month following report publication to allow the market time to process the disclosure. Positions were exited on the first trading day of the month after the subsequent quarterly report. In robustness checks, alternative exit rules were considered.

2.2 Feature Construction

Narratives were transformed into structured features using different types of large language models. Initially, experiments were conducted with multiple chat-based LLMs (*LLaMA-3-8B*, *LLaMA-4-Scout-17B*, *GPT-OSS-20B*, and *OpenAI-OSS-120B*) to classify narrative sentences into pre-defined financial and non-financial categories. Among these, *OpenAI-OSS-120B* was most effective in producing binary classifications. (Appendix I).

While this binary labeling captured category presence, it failed to convey how strongly the sentence was associated with each concept. To address this, we prompted the models to assign continuous scores between 0.0 and 1.0, indicating the relative strength of each category. However, when prompted this way (Appendix II), the models generated inconsistent numeric outputs, as the scores reflected token-level probabilities rather than true semantic proximity.

This limitation led to the adoption of an embedding-based cosine similarity approach (Appendix III), which enabled meaningful continuous representation of category relevance.

The following methods were used to collect scores on various categories:

- **Financial:** Sales, Expense, Profit, Operations, Liquidity, Investment, Financing, Litigation, Employment, Tax, Accounting (classified with embeddings, cosine threshold > 0.35).
- **Business:** Product, Buyer, Process, Location, Promotion, Supplier, Management, Strategy, Industry, Business Description (classified with embeddings).
- **Risk:** Firm-specific vs. Generic boilerplate (classified with MNLI LLM).
- **Analyst:** Detected if a sentence originated from an analyst or from management (MNLI).
- **Sentiment:** Negative, Neutral, Positive (FinBERT).
- **Subjective vs. Objective:** Rule-based using indicative words and numerical evidence.
- **Forward vs. Historical:** Rule-based using tense markers and modal verbs.

This produced a "narrative profile" for each firm-quarter, capturing financial and non-financial aspects. The differences between report features were calculated to measure shifts.

2.3 Moving Target Score (MTS)

Beyond variance- or frequency-weighted shifts, we experimented with distance-based and control theory based formulations to calculate moving targets. Among several candidates, two measures stood out for producing the most consistent signals: KL divergence and cosine distance. KL divergence penalizes changes in distributional weight across categories, while cosine distance emphasizes directional similarity in the feature space. The superior performance of these measures may stem from their ability to filter out noisy absolute magnitudes and instead isolate relative structural changes in firm communication.

$$MTS_{KL}(t) = \sum_{i=1}^{N} p_{i,t} \log\left(\frac{p_{i,t}}{p_{i,t-1}}\right), \tag{1}$$

where $p_{i,t}$ and $p_{i,t-1}$ are normalized feature weights in quarters t and $t-1$, respectively.

$$MTS_{\cos}(t) = 1 - \frac{\sum_{i=1}^{N} f_{i,t} f_{i,t-1}}{\sqrt{\sum_{i=1}^{N} f_{i,t}^2}\sqrt{\sum_{i=1}^{N} f_{i,t-1}^2}}, \tag{2}$$

Empirically, MTS_{KL} delivered a Sharpe ratio of 0.46 with 75% hit rate, while $MTS_{\cos}$ achieved an even stronger Sharpe of 0.54 with the same hit rate. Their relative success suggests that distance-based approaches, by capturing proportional and directional narrative shifts rather than absolute changes, provide a more robust mapping from text to returns.

2.4 Trading Strategy and Factor Analysis

Each quarter, firms were ranked by their MTS values. The top 10% were shorted and the bottom 10% were taken long (*flipped 10–10 portfolio*), with returns computed as the short–long spread. To evaluate whether these returns reflected true informational content rather than standard factor exposure, we regressed the portfolio returns on MKT, MOMENTUM, VALUE, and SIZE using OLS with Newey–West (1 lag) errors, and annualized quarterly alphas by a factor of four.

3 Data Insights

Several observations emerged from the dataset analysis:

- **Bang-Bang Max Delta:** Using the largest quarterly narrative shift variable as a predictor achieved 54.0% accuracy on up returns and 51.0% on down returns, suggesting extreme changes carry the most actionable information.
- **Distribution of Narrative Shifts:** On average, each firm's quarterly report exhibited changes in about 20 narrative features, far higher than the "2–3 shifts per quarter" intuition. This shows that narratives are highly dynamic from quarter to quarter, with many granular adjustments being captured in the deltas.
- **Financial vs. Non-Financial Shifts:** Nearly 9 out of 10 firm-quarters included at least one significant change in financial categories (e.g., Sales, Profit, Liquidity) and at least one change in business (non-financial) categories (e.g., Product, Management, Strategy). This indicates that both financial and non-financial framing are frequently adjusted, and shifts in non-financial narratives often preceded measurable financial deterioration, echoing patterns documented in the "Moving Targets" literature.
- **Risk Language:** Firm-specific risk mentions were sparse, while generic risk disclosures showed the strongest link to returns, with `delta_RiskGeneric_avg` correlated ≈ 0.14 with subsequent performance.
- **Timeframe Orientation:** Firms with above-median forward orientation in their disclosure earned an average return of 30.4% versus 7.1% for firms with lower forward orientation. This suggests a strong focus on future prospects is associated with substantially higher subsequent performance.
- **Sentence Structure (Subjective vs. Objective):** Firms in the top quartile of subjective language averaged a subsequent return of 50.8% compared with only 9.4% for the most objective firms. This suggests that opinion-heavy, narrative-driven disclosures were strongly associated with higher post-report performance in the post-2023 sample.

- **Analyst Interaction:** Sentences classified as analyst-originated accounted for only about 1.6% of the dataset, reflecting that LLMs are not very effective at detecting analyst questions.
- **Strategy Performance:** MTS variants showed mixed results. Baseline smoothers (rolling, EWMA, median) produced large negative returns, while weighted and ML approaches (variance, frequency, PCA, Lasso) offered little alpha. Distance-based measures—KL divergence and Cosine similarity—were more promising (Sharpe up to 0.54, hit rate 75%), though still factor-loaded and statistically weak.
- **Combined Strategy (MTS_combo):** To stabilize performance, a composite signal was constructed by combining the cosine similarity (MTS_cosine) and variance-weighted (MTS_w_var) approaches using a principal component analysis (PCA) method. The resulting portfolio delivered a Sharpe ratio of 0.48 with a cumulative return of 0.58 over eight quarters. This improvement suggests that PCA effectively extracts the common stable component across heterogeneous signals, dampening noise from individual predictors. By mixing a similarity-based measure (cosine) with a dispersion-sensitive weighting scheme (variance), the strategy balances directional strength with robustness, making it more resilient than either input alone.
- **Factor Exposures:** Factor regressions show that moving-average signals (rolling, EWMA, median) delivered large negative spreads with seemingly high alphas, but these were almost entirely explained by extreme loadings on momentum, value, and size. In contrast, variance- and frequency-weighted signals exhibited much weaker betas and negligible alpha, indicating closer alignment with standard factor returns. Distance-based signals such as KL divergence and Cosine produced more differentiated exposures, suggesting they capture narrative-driven structure not spanned by traditional factors, even though their alphas lacked statistical significance.

4 Conclusion

Data analysis revealed that narrative changes are frequent and multidimensional. Nearly every firm–quarter exhibited both financial and non-financial shifts, with risk framing and forward-looking orientation emerging as particularly influential. Extreme shifts (bang-bang deltas) were especially predictive, underscoring that large narrative adjustments carry the greatest informational value.

The exploration of multiple Moving Target Score (MTS) formulations revealed that while smoothing- and weighting-based signals (rolling, EWMA, variance, frequency) produced unstable or negligible returns, distance-based measures such as KL divergence and cosine similarity offered more consistent predictive content. These measures filtered out noise in absolute magnitudes and instead highlighted proportional and directional narrative re-alignments that markets tend to underreact to.

While LLM-based classification captured broad narrative dimensions effectively, it was less capable of reliably detecting analyst-originated statements or subtle dialogue nuances in transcripts. This suggests that hybrid approaches, blending rule-based heuristics and advanced models, are needed to better capture fine-grained interactions.

Appendix I — Discrete Classification (LLM Prompt and Output)

```
    prompt = f"""
Classify the following sentence into categories.
Each category should be 0 (not present) or 1 (present).
Sentiment can be -1 (negative), 0 (neutral), or +1 (positive).

Base your assessment only on the documents provided and not your training knowledge of the company, industry or market.

Categories:
Financial: Sales/Revenue, Cost/Expense, Profit/Loss, Operations, Liquidity, Investment, Financing, Litigation, Employment, Regulation/Tax,
Accounting
Non-financial: Product, People (buyer), Process, Location, Macro environment, Promotion, People (seller), Management
Risk related: Firm specific risk, Generic boilerplate
Objective vs Subjective: Objective, Subjective
Timeframe: Forward looking, Historical
Sentiment Score: -1, 0, +1

Return JSON like:
{{
 "Sales/Revenue": 1, "Cost/Expense": 0, ..., "Sentiment": +1
}}

Sentence: "{sentence}"
"""
```

Figure 1: Prompt used to instruct chat-based LLMs for discrete 0/1 category classification.

```
Block path: data/reports/INSULET_CORP/f_goml27vSnN4MeqhKpqVWnM_blocks.txt
Sentence: Capital Spending—Capital expenditures were $44.6 million and $26.2 million for the six months ended June 30, 2024 and 2023, resp
ectively, and
primarily related to the purchase of equipment to increase our manufacturing capacity. We expect capital expenditures for 20
24 to increase
compared with 2023 given the timing of spending on machinery, equipment and tooling for our new Malaysia manufacturing facility and to suppor
t
continuous improvement efforts in our other manufacturing locations. To a lesser extent, we expect capital expenditures to in
crease due to
investments in our information technology infrastructure. We expect to fund our capital expenditures using existing cash.

Model: gpt-oss-120b
{'Sales/Revenue': 0, 'Cost/Expense': 0, 'Profit/Loss': 0, 'Operations': 0, 'Liquidity': 1, 'Investment': 1, 'Financing': 0, 'Litigation': 0,
'Employment': 0, 'Regulation/Tax': 0, 'Accounting': 0, 'Product': 0, 'People (buyer)': 0, 'Process': 0, 'Location': 1, 'Macro environment':
0, 'Promotion': 0, 'People (seller)': 0, 'Management': 0, 'Firm specific risk': 0, 'Generic boilerplate': 1, 'Objective': 1, 'Subjective': 0,
'Forward looking': 1, 'Historical': 1, 'Sentiment': 0}

Model: gpt-oss-20b
{'Error': '```json\n{\n  "Sales/Revenue": 0,\n  "Cost/Expense": 1,\n  "Profit/Loss": 0,\n  "Operations": 1,\n  "Liquidity": 1,\n  "Investmen
t": 1,\n  "Financing": 0,\n  "Litigation": 0,\n  "Employment": 0,\n  "Regulation/Tax": 0,\n  "Accounting": 0,\n  "Product": 0,\n  "People (bu
yer)": 0,\n  "Process": 1,\n  "Location": 1,\n  "Macro environment": 0,\n  "Promotion": 0,\n  "People (seller)": 0,\n  "Management": 0,\n  "F
irm specific risk": 0,\n  "Generic boilerplate": 1,\n  "Objective": 1,\n  "Subjective": 0,\n  "Forward looking": 1,\n  "Historical": 1,\n  "S
entiment": 0\n}\n```'}

Model: llama-3.1-8b
{'Error': 'Here\'s a classification of the given sentence into the provided categories:\n\n{\n  "Financial": \n    "Capital Spending": 1,\n
"Cost/Expense": 0,\n    "Profit/Loss": 0,\n    "Operations": 1,\n    "Liquidity": 1,\n    "Investment": 1, \n    "Financing": 1, \n    "Regul
ation/Tax": 0,\n    "Accounting": 0,\n  "Non-financial": \n    "Product": 0,\n    "People (buyer)": 0,\n    "Process": 1,\n    "Location":
1,\n    "Macro environment": 0,\n    "Promotion": 0,\n    "People (seller)": 0,\n    "Management": 0,\n  "Risk related": \n    "Firm specific
risk": 0,\n    "Generic boilerplate": 0,\n  "Objective vs Subjective": \n    "Objective": 1,\n    "Subjective": 0,\n  "Timeframe": \n    "For
ward looking": 1,\n    "Historical": 0,\n  "Sentiment": \n    "-1": 0,\n    "0": 0,\n    "+1": 1\n}\n\nBased on the classification:\n\n* The
financial categories present are Capital Spending, Operations, Liquidity, Investment, and Financing.\n* The non-financial categories present
are Operations, Process, and Location.\n* There are no risk-related categories present.\n* The statement is objective.\n* The statement has a
forward-looking timeframe.\n* The sentiment score is +1 (positive).'}

Model: llama-4-scout-17b
{'Error': 'Here is the classification of the sentence into the given categories:\n\n```\n{\n  "Financial": {\n    "Sales/Revenue": 0,\n    "C
ost/Expense": 1,\n    "Profit/Loss": 0,\n    "Operations": 1,\n    "Liquidity": 0,\n    "Investment": 1,\n    "Financing": 0,\n    "Litigatio
n": 0,\n    "Employment": 0,\n    "Regulation/Tax": 0,\n    "Accounting": 1\n  },\n  "Non-financial": {\n    "Product": 0,\n    "People (buye
r)": 0,\n    "Process": 0,\n    "Location": 1,\n    "Macro environment": 0,\n    "Promotion": 0,\n    "People (seller)": 0,\n    "Managemen
t": 0\n  },\n  "Risk related": {\n    "Firm specific risk": 0,\n    "Generic boilerplate": 0\n  },\n  "Objective vs Subjective": {\n    "Obje
ctive": 1,\n    "Subjective": 0\n  },\n  "Timeframe": {\n    "Forward looking": 1,\n    "Historical": 1\n  },\n  "Sentiment Score": 0\n}\n```
\n\nHere\'s a brief explanation of the classification:\n\n* Financial:\n\t+ Cost/Expense: 1 (capital expenditures mentioned)\n\t+ Operations:
1 (manufacturing capacity, equipment purchase)\n\t+ Investment: 1 (investments in IT infrastructure, machinery, equipment, and tooling)\n\t+
Accounting: 1 (capital expenditures, funding using existing cash)\n* Non-financial:\n\t+ Location: 1 (Malaysia manufacturing facility)\n* Ris
k related: None present\n* Objective vs Subjective: Objective (factual statements about capital expenditures)\n* Timeframe:\n\t+ Forward look
ing: 1 (expectations for 2024)\n\t+ Historical: 1 (capital expenditures for 2023 and 2024 mentioned)\n* Sentiment Score: 0 (neutral tone, no
positive or negative emotions expressed)'}
```

Figure 2: Example discrete classification output from LLMs showing binary (0/1) category assignments.

Appendix II — Continuous Scoring Attempt (LLM Prompt and Output)

```
    prompt = f"""
You are an analyst evaluating corporate disclosures.

Classify the following sentence into the categories below.
Assign a CONTINUOUS score between 0.0 and 1.0 to represent
how strongly each category is expressed in the sentence
(0 = not mentioned at all, 1 = very strongly emphasized).

Sentiment can take any value between -1.0 (very negative) and +1.0 (very positive).

Base your assessment only on the documents provided and not your training knowledge of the company, industry or market.

Categories:
Financial: Sales/Revenue, Expense/Cost, Profit/Loss, Operations, Liquidity, Investment, Financing, Litigation, Employment, Regulation/Tax,
Accounting
Non-financial: Product, Buyer/Customer, Process/Technology, Location/Market, Promotion, Supplier/Partner, Management, Strategy,
Industry/Structure
Risk-related: Firm-specific Risk, Generic Risk
Other dimensions: Objective, Subjective, Forward-looking, Historical
Sentiment Score: range -1.0 to +1.0

Return valid JSON, e.g.:
{{
 "Sales/Revenue": 0.45,
 "Expense/Cost": 0.12,
 "Investment": 0.76,
 "Operations": 0.51,
 "Liquidity": 0.39,
 "Sentiment": 0.1
}}

Sentence: "{sentence}"
"""
```

Figure 3: Modified prompt instructing LLMs to assign continuous scores (0.0–1.0) to indicate the strength of each category in the sentence.

```
📘 Sentence:
Capital Spending—Capital expenditures were $44.6 million and $26.2 million for the six months ended June 30, 2024 and 2023, respectively, and
primarily  related  to  the  purchase  of  equipment  to  increase  our  manufacturing  capacity.  We  expect  capital  expenditures  for  20
24  to  increase compared with 2023 given the timing of spending on machinery, equipment and tooling for our new Malaysia manufacturing facil
ity and to support continuous  improvement  efforts  in  our  other  manufacturing  locations.  To  a  lesser  extent,  we  expect  capital  e
xpenditures  to  increase  due  to investments in our information technology infrastructure. We expect to fund our capital expenditures using
existing cash.

==============================
🤖 Model: openai-120b
{
  "Sales/Revenue": 0.0,
  "Expense/Cost": 0.15,
  "Profit/Loss": 0.0,
  "Operations": 0.4,
  "Liquidity": 0.5,
  "Investment": 0.9,
  "Financing": 0.3,
  "Litigation": 0.0,
  "Employment": 0.0,
  "Regulation/Tax": 0.0,
  "Accounting": 0.1,
```

Figure 4: Example output of OpenAI-120B attempting continuous scoring. The scores are inconsistent and not semantically grounded.

Appendix III — Embedding-Based Cosine Similarity

```
📘 Sentence:
Capital Spending—Capital expenditures were $44.6 million and $26.2 million for the six months ended June 30, 2024 and 2023, respectively, and
primarily related to the purchase of equipment to increase our manufacturing capacity. We expect capital expenditures for 20
24 to increase compared with 2023 given the timing of spending on machinery, equipment and tooling for our new Malaysia manufacturing facil
ity and to support continuous improvement efforts in our other manufacturing locations. To a lesser extent, we expect capital e
xpenditures to increase due to investments in our information technology infrastructure. We expect to fund our capital expenditures using
existing cash.

=== Finance Category Similarities ===
Sales/Revenue        | avg: 0.236 | max: 0.313
Expense/Cost         | avg: 0.286 | max: 0.365 ✅ (above threshold)
Profit/Loss          | avg: 0.239 | max: 0.297
Operations           | avg: 0.274 | max: 0.421 ✅ (above threshold)
Liquidity            | avg: 0.282 | max: 0.371 ✅ (above threshold)
Investment           | avg: 0.331 | max: 0.536 ✅ (above threshold)
Financing            | avg: 0.269 | max: 0.410 ✅ (above threshold)
Litigation           | avg: 0.123 | max: 0.226
Employment           | avg: 0.193 | max: 0.278
Regulation/Tax       | avg: 0.172 | max: 0.262
Accounting           | avg: 0.233 | max: 0.336
```

Figure 5: Embedding-based similarity results using OpenAI `text-embedding-3-small`. Cosine similarity provides continuous, interpretable relationships across multiple categories (e.g., Cost, Investment, Financing).